

RESEARCH PORTFOLIO PAPER

Computational Analysis of Spanish-English Code-switching

A corpus-based study of educational variation in the Bangor Miami Corpus

Python | Corpus Linguistics | Bilingual Language Data

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Abstract

This study presents a reproducible computational analysis of Spanish-English code-switching in the Bangor Miami Corpus. It asks whether the ordinal education codes supplied with the corpus are associated with participant-level switching frequency. A Python pipeline was developed to parse CHAT headers, align corpus speaker codes with questionnaire records, read heterogeneous word-level TSV files, and identify conservative within-utterance language transitions. To avoid unsupported identity assignments, a speaker mapping was confirmed only when recording filename, age at recording, and sex uniquely identified one questionnaire row. The final strict cohort contained 73 independent participants and 195,864 valid English/Spanish tokens. A switch was counted only when adjacent valid tokens changed from English to Spanish or Spanish to English within the same recording, speaker, and utterance; other labels reset the sequence. The pipeline detected 2,980 switches, or 15.21 per 1,000 valid tokens. Mean participant rates were 13.63, 14.81, and 15.74 in the researcher-defined Low, Middle, and High education groups. Welch ANOVA, Kruskal-Wallis analysis, ordinary ANOVA, and HC3-robust ordinal regression provided no reliable evidence of an education effect. The study therefore contributes an auditable corpus-processing workflow rather than a positive causal claim.

Keywords: code-switching; computational linguistics; Spanish-English bilingualism; corpus analysis; reproducible research

1. Introduction

Code-switching is a central feature of bilingual communication in which speakers draw on more than one linguistic system within an interaction. Research has examined its grammatical organisation, discourse functions, and relationship with bilingual identities and communities (Gumperz, 1982; Poplack, 1980; Myers-Scotton, 1993; Zentella, 1997). However, a sociolinguistic question cannot be answered reliably by counting visually salient bilingual examples alone. Corpus files must be aligned with participant metadata, the analytical unit must be defined, and ambiguous language labels must be handled consistently.

The original proposal for this project hypothesised that educational level might influence the frequency and structural preferences of Spanish-English code-switching among Miami bilinguals. The computational audit showed that a narrower first study was necessary. The transcript and questionnaire files use different identifier systems, education is supplied as an unexplained five-level ordinal code, and the corpus contains mixed and non-lexical language labels. Consequently, the present portfolio focuses on one reproducible outcome: conservative within-utterance English-Spanish transitions per 1,000 valid tokens.

The study addresses two questions. First, can a transparent Python workflow link the supplied corpus and questionnaire files without guessing uncertain speaker identities? Second, do participant-level switching rates differ across three researcher-defined ordinal education groups? The emphasis is on operational clarity, quality control, and reproducibility.

2. Background

Poplack's (1980) typological work distinguished switching patterns according to their structural location and helped establish code-switching as systematic bilingual behaviour rather than random interference. Gumperz (1982) foregrounded its conversational and contextual functions, while Myers-Scotton (1993) developed a grammatical account of asymmetries between participating languages. Ethnographic work such as Zentella (1997) further demonstrated that bilingual choices are embedded in community practices and social identities.

These traditions caution against reducing code-switching to a single social cause. Education may correlate with language experience, literacy, occupation, network composition, and interactional setting, but a corpus association would not establish causality. The present analysis therefore treats education as an exploratory speaker-level predictor and retains age, proficiency, topic, and interlocutor effects as limitations rather than inventing controls

that the verified pipeline has not yet modelled.

The Bangor Miami Corpus is valuable because it contains naturally occurring Spanish-English conversations, CHAT transcripts, word-level annotations, and questionnaire metadata (Deuchar et al., 2014). At the same time, its structure creates a genuine computational-linguistics task: transcript speaker codes must be reconciled with separate questionnaire identifiers, and language sequences must be derived from annotated token rows while respecting utterance boundaries.

3. Data and Methods

3.1 Data audit and ethical boundary

The downloaded corpus comprised 56 CHAT files and 56 word-level TSV files. The TSV collection contained 316,600 analyzable rows after non-token footer records were removed. The public GitHub repository does not redistribute corpus transcripts, questionnaire responses, audio, or full token exports. It contains modular code, synthetic fixtures, aggregate results, and instructions for users who obtain the corpus through its official provider.

3.2 Conservative speaker alignment

CHAT/TSV speaker codes and questionnaire IDs were treated as distinct systems. Each CHAT header was parsed for recording, speaker code, participant role, age, and sex. A mapping was labelled confirmed only when exactly one questionnaire row shared the normalised recording filename, age at recording, and normalised sex. Filename-only and elimination-based assignments remained unresolved. This rule produced 74 confirmed recording-speaker mappings representing 73 independent participants. Unresolved cases were preserved in an audit table and excluded from education analysis.

3.3 Education recoding

The questionnaire contained ordinal education levels 1 to 5 but did not provide verified descriptive labels for those codes. The original values were therefore retained and adjacent levels were recoded solely for group comparison: Low (1-2), Middle (3), and High (4-5). These labels describe ordinal position, not primary, secondary, university, or postgraduate attainment. The strict sample contained 5, 31, and 37 participants in the three groups.

3.4 Operational definition of switching

Tokens were ordered within recording, speaker, and utterance. A switch event was recorded only for adjacent eng-to-spa or spa-to-eng transitions. Labels such as 999, mixed-language forms, and transcription artefacts reset the sequence, so the detector did not bridge across ambiguous material. It also never joined the last token of one utterance to the first token of another. Participant rate was calculated as switch count divided by valid eng/spa tokens, multiplied by 1,000.

3.5 Statistical analysis

Group means, standard deviations, and medians were calculated at participant level. Welch one-way ANOVA was the preferred exploratory comparison because group sizes were highly unequal. Kruskal-Wallis analysis served as a non-parametric sensitivity check, and ordinary one-way ANOVA was retained as the proposal-aligned supplementary test. A Brown-Forsythe test assessed variance differences. The original five-level education code was also entered as an ordinal predictor in a linear regression with HC3 robust standard errors. IQR-defined outliers were excluded only in a declared sensitivity analysis. No post-hoc pairwise tests were performed because no omnibus result was significant.

4. Results

The strict cohort produced 195,864 valid English/Spanish tokens and 2,980 conservative switch events: 1,293 English-to-Spanish and 1,687 Spanish-to-English transitions. The pooled rate was 15.21 events per 1,000 valid tokens.

Education group	n	Mean rate	SD	Median
Low (levels 1-2)	5	13.63	11.05	9.51
Middle (level 3)	31	14.81	11.63	12.76
High (levels 4-5)	37	15.74	10.47	14.14

Table 1. Participant-level conservative switch rates per 1,000 valid tokens.

Welch ANOVA found no reliable group difference, $F(2, 11.31) = 0.109$, $p = .8976$. Kruskal-Wallis analysis was consistent, $H(2) = 0.341$, $p = .8434$, as was ordinary ANOVA, $F(2, 70) = 0.114$, $p = .8927$, eta-squared = .0032. Brown-Forsythe testing did not indicate unequal group variances, $p = .6823$. In the ordinal regression, each one-level increase in education was associated with an estimated 0.639 additional switches per 1,000 tokens, $HC3\ SE = 1.325$, $t(71) = 0.483$, $p = .6309$, 95% normal-approximation CI [-1.96, 3.24], $R\text{-squared} = .0028$.

Two participant rates were flagged by the within-group IQR rule. Excluding them did not reverse the conclusion: Welch $p = .2180$ and Kruskal-Wallis $p = .4541$. The visually increasing means therefore remain descriptive rather than statistically supported.

5. Discussion

The results do not support the original prediction that educational level reliably differentiates conservative Spanish-English switching frequency. More importantly, they demonstrate why a computational portfolio should not be judged by whether it obtains a significant p-value. The strongest contribution is the auditable chain from raw corpus structure to participant-level evidence: identifiers are aligned conservatively, language boundaries are explicit, and every exclusion is retained for review.

Several factors may explain the null finding. The Low group contains only five participants, which produces imprecise estimates despite its apparently lower mean. Education codes may also be too coarse to capture bilingual experience. Naturally occurring conversations vary by topic, interlocutor relationship, recording length, and local language norms. Finally, the conservative metric excludes transitions separated by mixed or ambiguous tokens and does not capture discourse-level or inter-sentential switching. It is therefore a precise measure of one phenomenon, not a complete measure of bilingual language behaviour.

The analysis also refines the project's application narrative. Rather than claiming a transition from languages into generic artificial intelligence, it shows how linguistic training guides computational decisions. Knowledge of utterance structure, bilingual annotation, and sociolinguistic interpretation is necessary to decide what the code should count, reset, merge, or leave unresolved. This combination is directly relevant to computational linguistics programmes concerned with multilingual and variable language data.

6. Limitations and Future Work

The strict mapping policy reduces the available sample but protects validity. The education grouping is researcher-defined and unbalanced. Rates treat participants as independent while conversations may share interlocutors and contexts. The present regression models rate rather than a count distribution with token exposure, and it does not include verified proficiency, topic, age, or gender controls. These limitations prevent causal interpretation.

A next statistical extension could model switch counts with exposure and conversation-level clustering. A linguistic extension could distinguish intra-sentential, inter-sentential, and mixed-morpheme phenomena using manually reviewed annotations. A machine-learning extension could predict token-level switch points using speaker-disjoint cross-validation. Such prediction should be framed as an extension, not retrospectively claimed as part of the present project.

7. Conclusion

This study transformed a sociolinguistic proposal into a reproducible Python workflow for annotated bilingual speech. The pipeline parsed corpus formats, audited speaker metadata alignment, operationalised conservative switching, generated participant rates, and reproduced multiple inferential checks. Within the strictly matched cohort, education level did not reliably explain conservative switching frequency. The null result is retained because transparent negative evidence is more valuable than an unsupported positive claim. As a portfolio, the project demonstrates corpus reasoning, multilingual data processing, statistical caution, and research reproducibility.

References

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